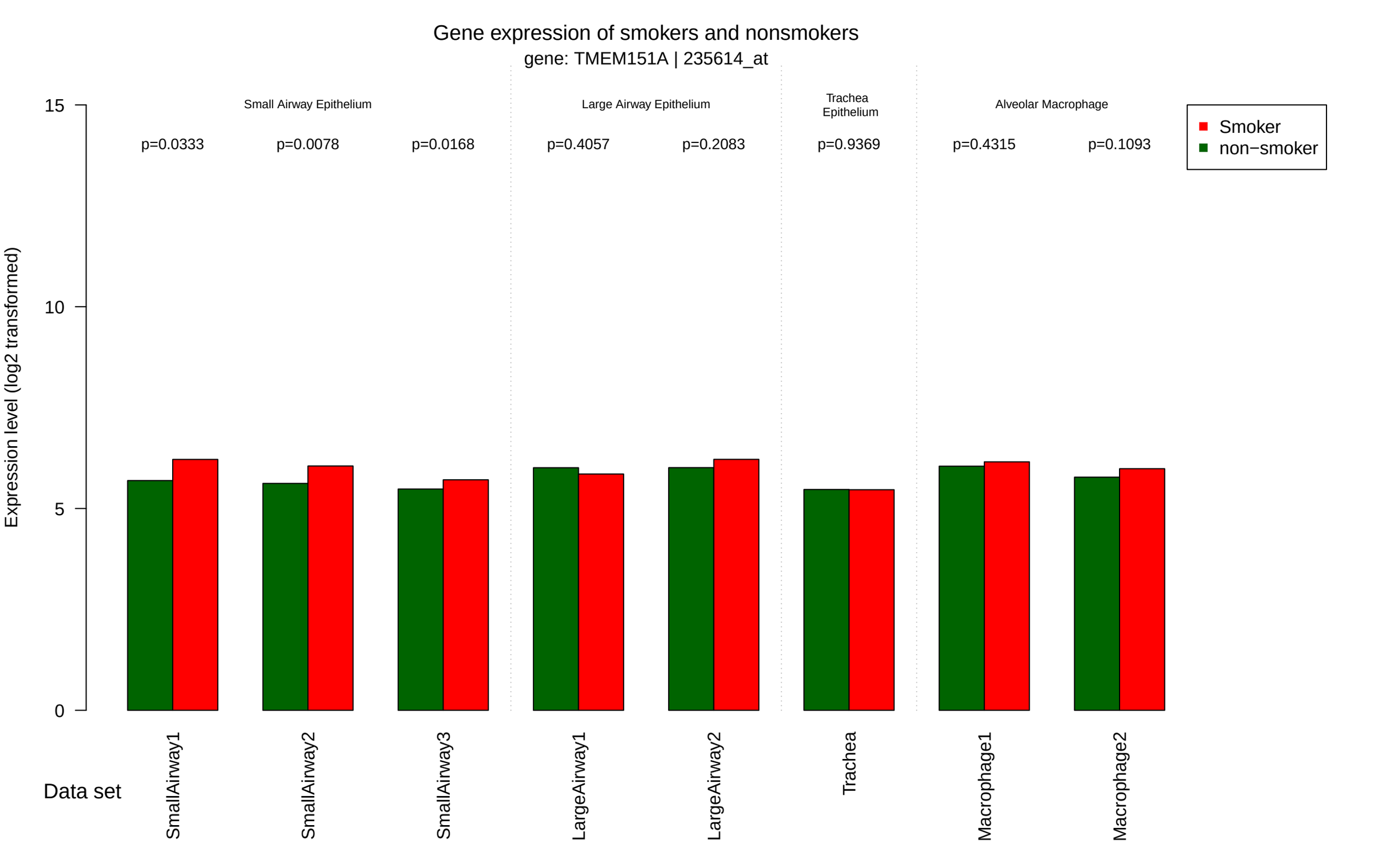
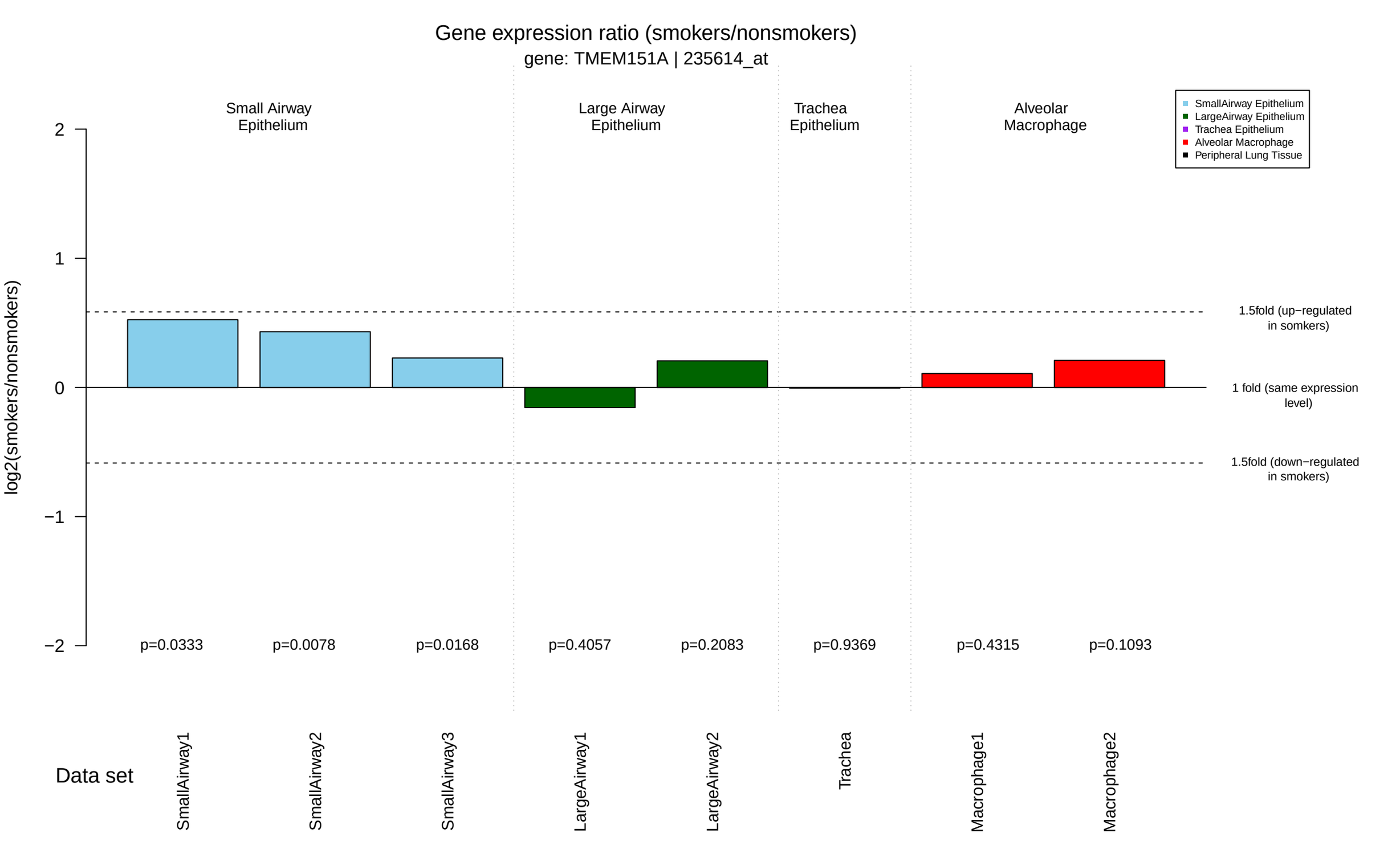
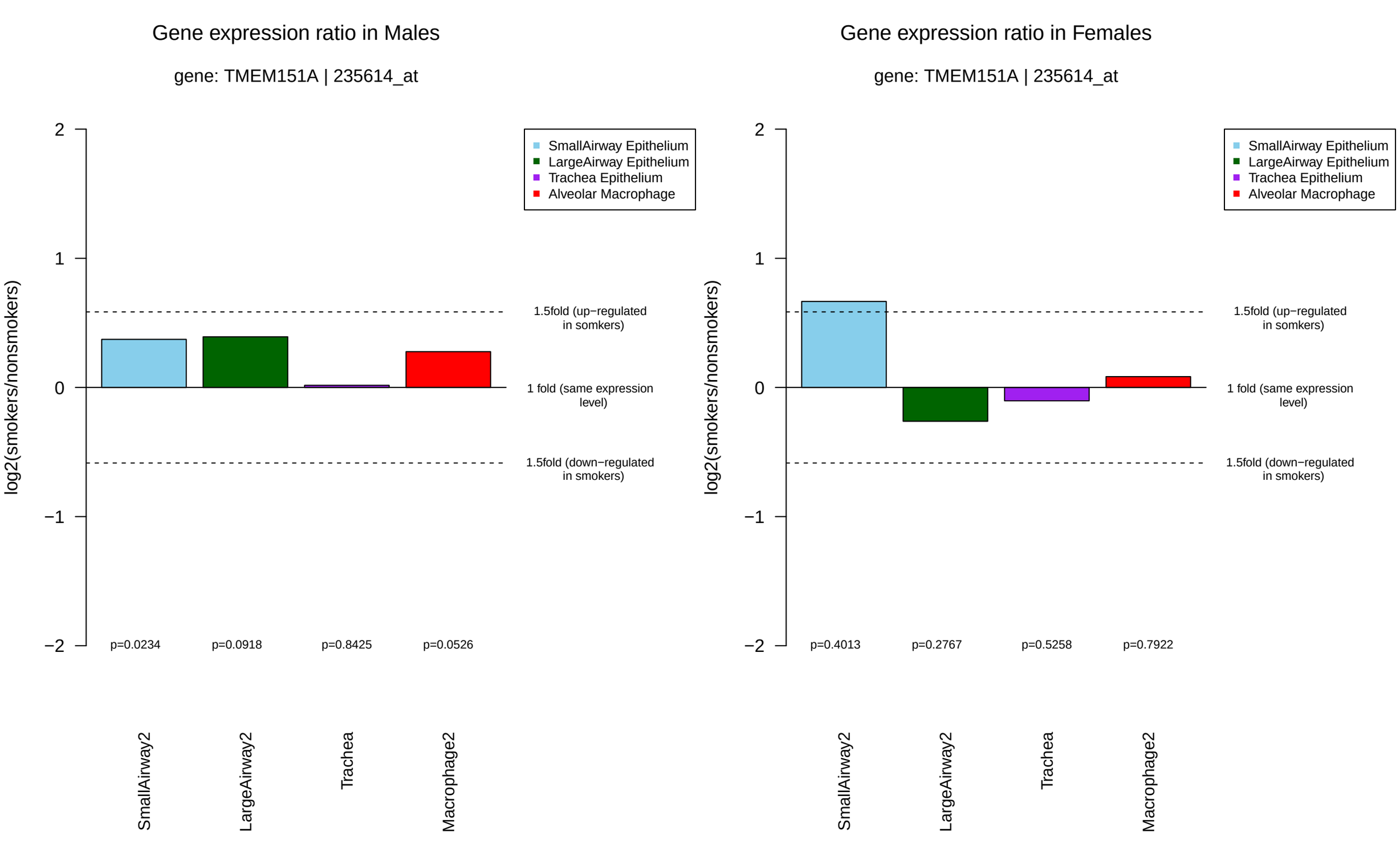
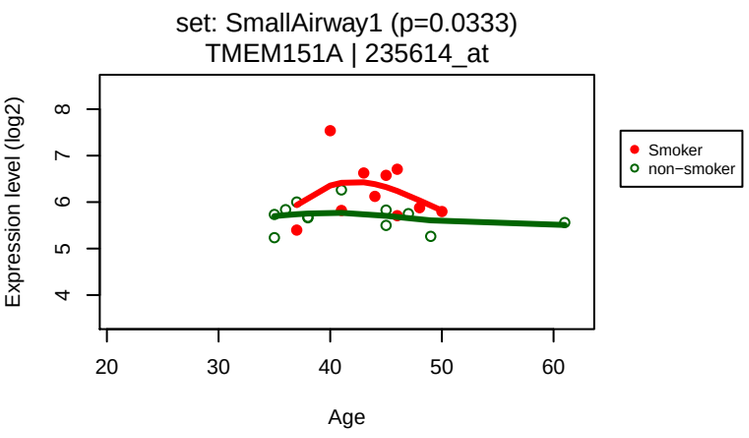
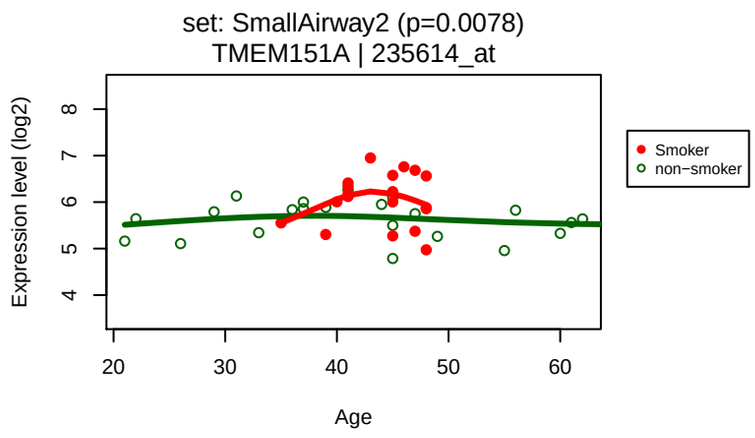
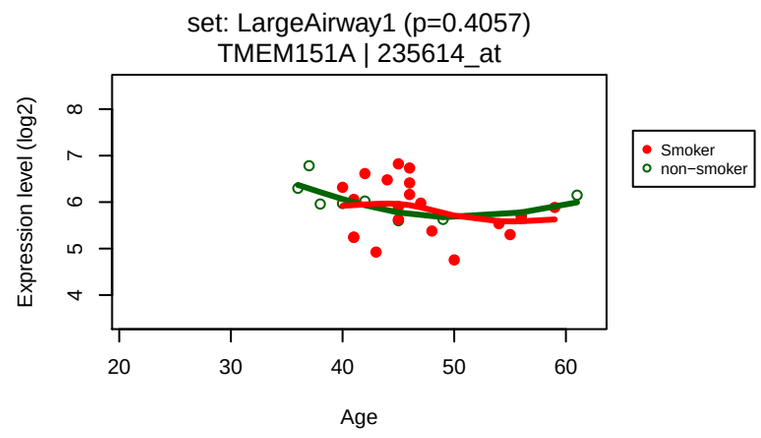

Gene expression of smokers and nonsmokers
gene: TMEM151A | 235614_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0333
p=0.0078
p=0.0168
p=0.4057
p=0.2083
p=0.9369
p=0.4315
p=0.1093
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

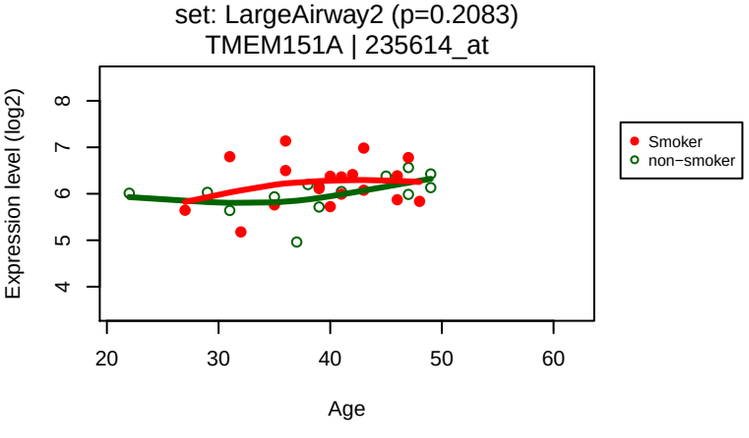

Gene expression ratio (smokers/nonsmokers)
gene: TMEM151A | 235614_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0333
p=0.0078
p=0.0168
p=0.4057
p=0.2083
p=0.9369
p=0.4315
p=0.1093
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

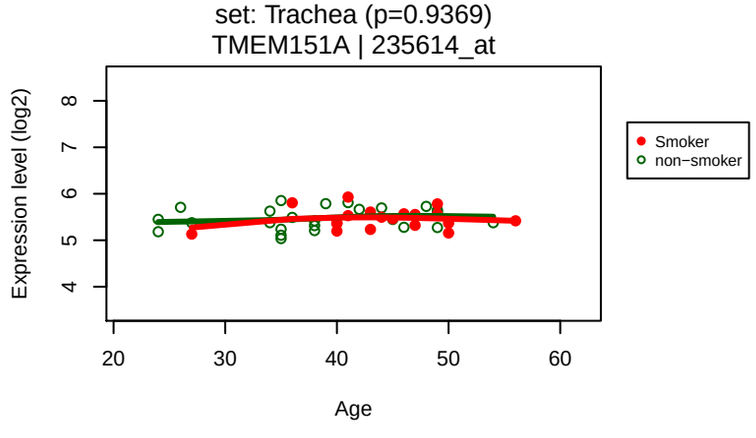

Gene expression ratio in Males
gene: TMEM151A | 235614_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0234
p=0.0918
p=0.8425
p=0.0526
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM151A | 235614_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4013
p=0.2767
p=0.5258
p=0.7922
SmallAirway2
LargeAirway2
Trachea
Macrophage2

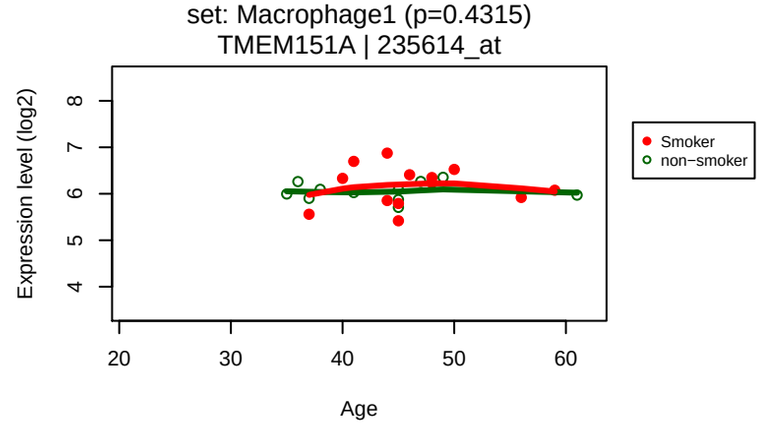

set: SmallAirway1 (p=0.0333)
TMEM151A | 235614_at
Expression level (log2)
Age
Smoker
non−smoker

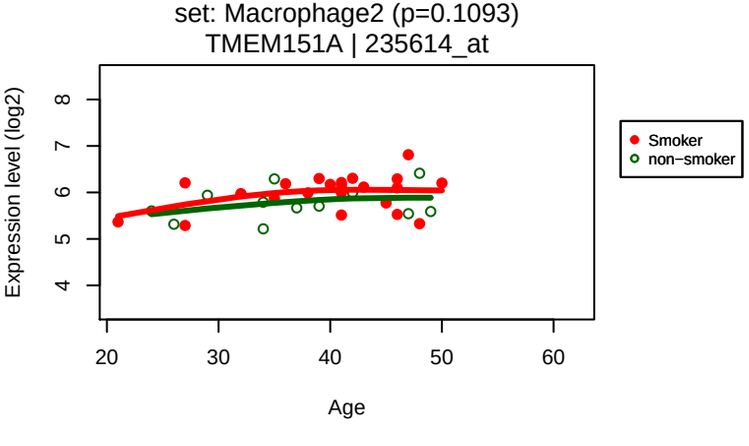

set: SmallAirway2 (p=0.0078)
TMEM151A | 235614_at
Expression level (log2)
Age
Smoker
non−smoker

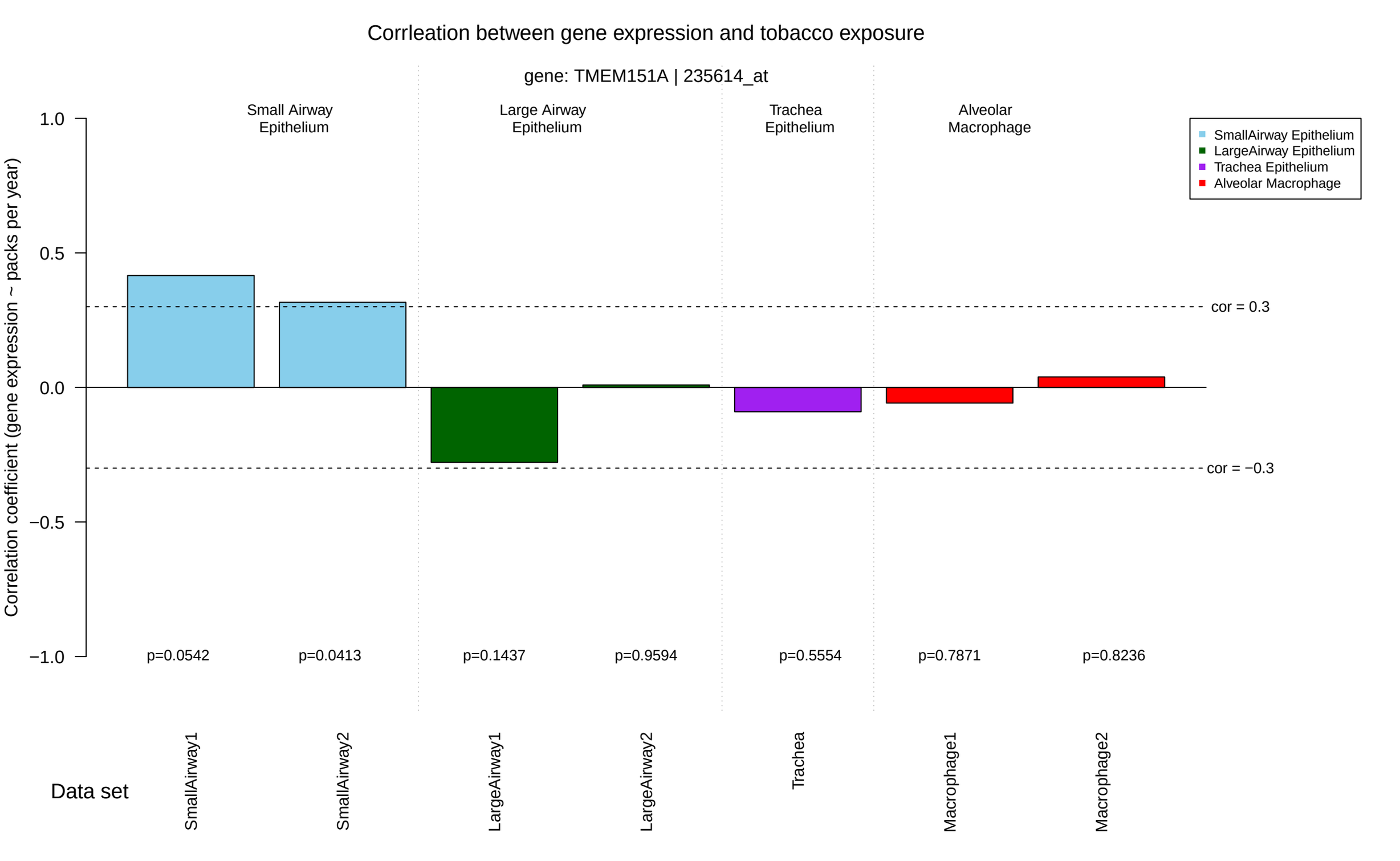

set: LargeAirway1 (p=0.4057)
TMEM151A | 235614_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2083)
TMEM151A | 235614_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9369)
TMEM151A | 235614_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4315)
TMEM151A | 235614_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1093)
TMEM151A | 235614_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM151A | 235614_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0542
p=0.0413
p=0.1437
p=0.9594
p=0.5554
p=0.7871
p=0.8236
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2